Crypto Market Dynamics: Evidence Report

Acquisition cutoff: 2026-06-30
Estimator seed: 20260713
Scope: descriptive and associational empirical finance; no forecasting, trading strategy, portfolio-allocation, or causal claim.

Study Design

The repository addresses four questions: realized crypto dependence and TradFi integration; institutional market plumbing; leverage, tails, and connectedness; and endogenous liquidity state with monthly point-in-time market structure. Inputs, timestamps, availability rules, transformations, denominators, missing-value rules, samples, estimands, and claims are registered under `research/`.

S1 covers BTC/ETH daily anchors. S2 fixes a January 2021 PIT-eligible stable core with at least 95% daily coverage. S3 is a supplementary unbalanced current-cohort panel and remains survivorship-biased. S4 is monthly point-in-time top-100 structure through the last complete month. S5 begins each ETF or institutional series at its actual reporting inception.

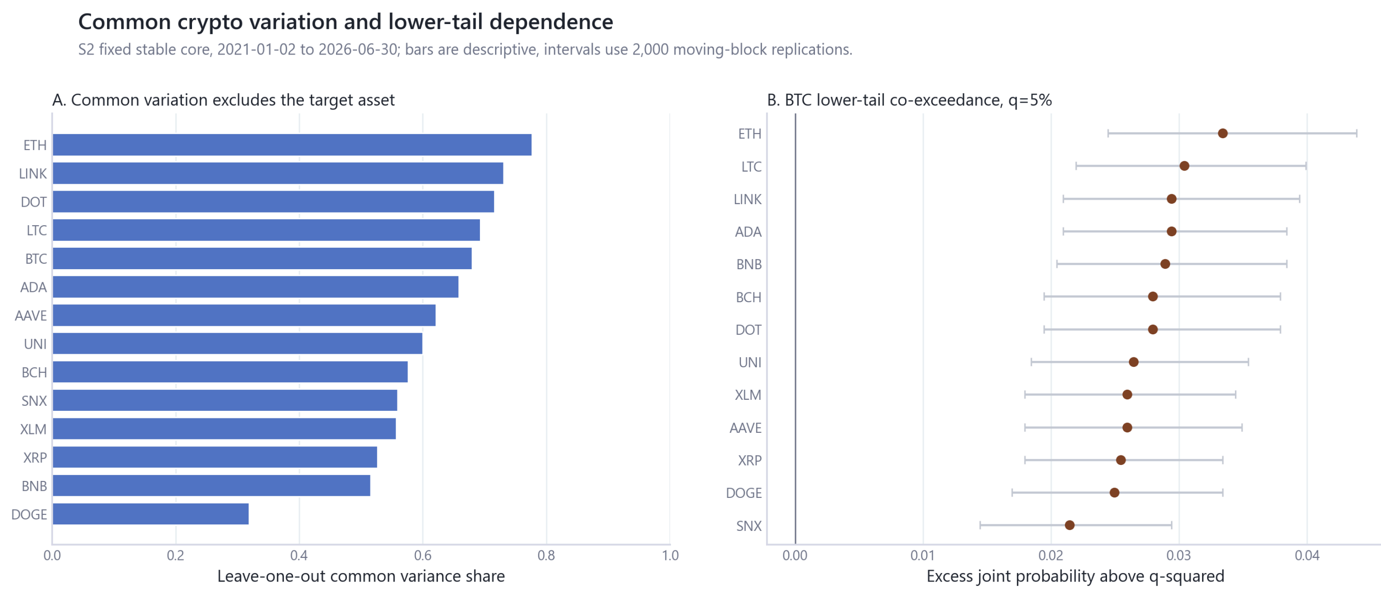
The primary uncertainty procedures are HAC covariance and deterministic moving-block bootstrap inference. Same-day MVRV is excluded from primary BTC/ETH models. ETF flows, stablecoin supply, and DeFi TVL are endogenous market-state measures.

Estimands and Equations

Quantity	Definition	Interpretation boundary
Log return	`r[i,t] = log(P[i,t] / P[i,t-1])` on the series' native calendar before joins	Realized return, not a forecast target.
Leave-one-out common factor	First PC of standardized S2 returns excluding asset `i`; `r[i,t] = alpha + beta * PC1[-i,t] + error`	Descriptive common variation without mechanical self-inclusion.
Tail excess	`Pr(r[i] <= Q[i,q], r[j] <= Q[j,q]) - q^2`	Co-exceedance relative to independence, not causal contagion.
ETF distributed lag	`y[t] = alpha + sum(k=0..5) beta[k] * flow_bps[t-k] + error[t]`	Timing-sensitive association; flow uses lagged market capitalization.
Leverage tail model	`logit Pr(r[t+1] <= Q[0.05]) = spline(leverage[t-1]) + spline(volatility[t-1])`	Conditional association, not a prediction rule.
Generalized FEVD connectedness	Off-diagonal generalized forecast-error variance shares divided by total shares	Descriptive stress connectedness; variable-order sensitivity is reported.

Quantity	Definition	Interpretation boundary
PIT structure	`HHI[t] = sum(i) share[i,t]^2`; effective count `= 1 / HHI[t]`; turnover `= 0.5 * sum(i) abs(share[i,t] - share[i,t-1])`	Monthly composition only; no daily constituent-return inference.
Liquidity residual	Residual from HAC regression of log USD TVL growth on BTC, ETH, and TOTAL3 returns	Endogenous state proxy, not an exogenous liquidity shock.
MVRV identity	`d log MVRV = d log market cap - d log realized cap + residual`	Measurement-mechanics diagnostic excluded from primary return models.

01. Cross-Asset Dependence and Regimes



Result. Within S2, full-system PC1 accounts for 66.0% of standardized return variation, while median 5% lower-tail co-exceedance is 2.8% above the independence benchmark.

Sample. 2021-01-02 to 2026-06-30, n=2006

Method and uncertainty. Leave-one-out PCA/HAC factor regressions and moving-block tail inference. Intervals, diagnostics, and sensitivity specifications are reported in the linked source tables.

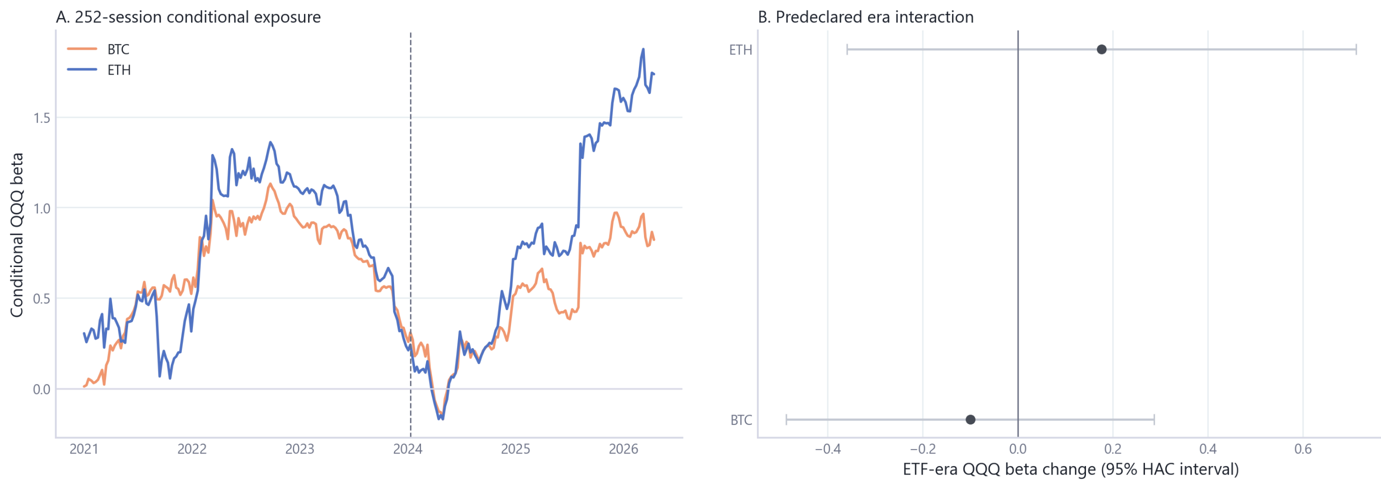
Evidence grade. B. **Limitation.** S2 is a fixed 2021 PIT-eligible universe; results do not establish causation, forecasting value, or investability.

Provenance. [Primary source table](); [module documentation]().

02. Macro and TradFi Integration

Dynamic TradFi integration

Crypto returns are joined to native-session TradFi returns after calculation; contemporaneous association only.



Result. Conditional QQQ exposure estimates differ across the predeclared BTC ETF-era split: BTC 1.03 to 0.93 (change p=0.611); ETH 1.33 to 1.51 (change p=0.520).

Sample. 2020-01-03 to 2026-04-14, n=1577

Method and uncertainty. 252-session multivariate HAC exposures and formal era interactions. Intervals, diagnostics, and sensitivity specifications are reported in the linked source tables.

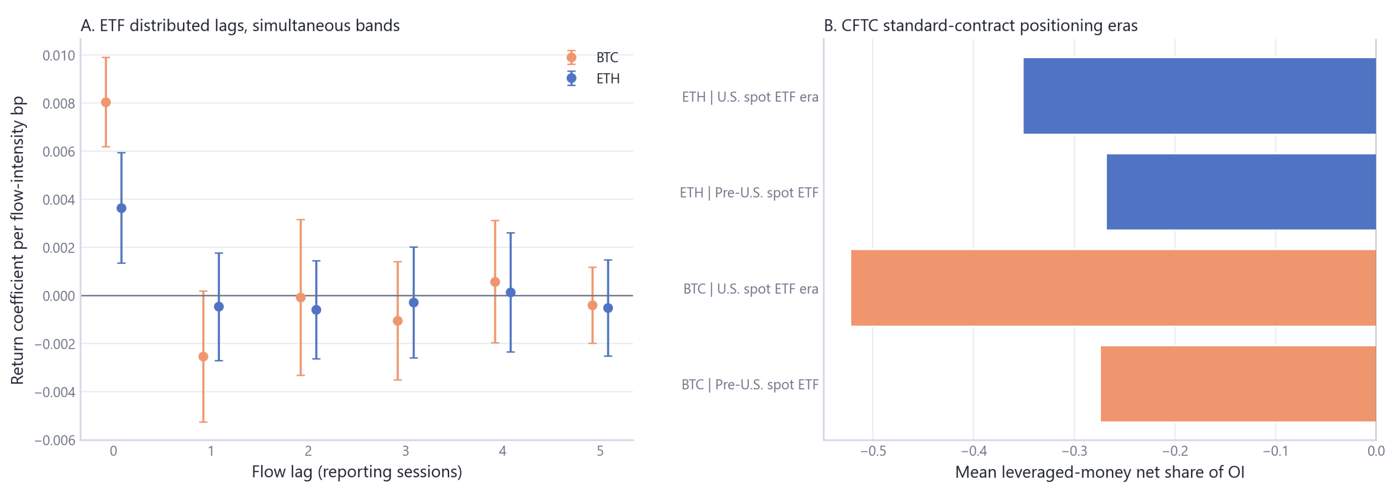
Evidence grade. B. **Limitation.** The split is descriptive; contemporaneous exposures do not identify ETF effects or causal transmission.

Provenance. Primary source table; module documentation.

04. ETF and Institutional Flows

ETF flows and regulated-futures positioning

ETF timing remains unresolved; CFTC Tuesday positions use a Friday availability proxy and are separate descriptive context.



Result. Across 12 asset-lag return coefficients, 2 have 95% moving-block simultaneous intervals excluding zero under the reported-date convention. Simultaneity prevents a price-impact interpretation.

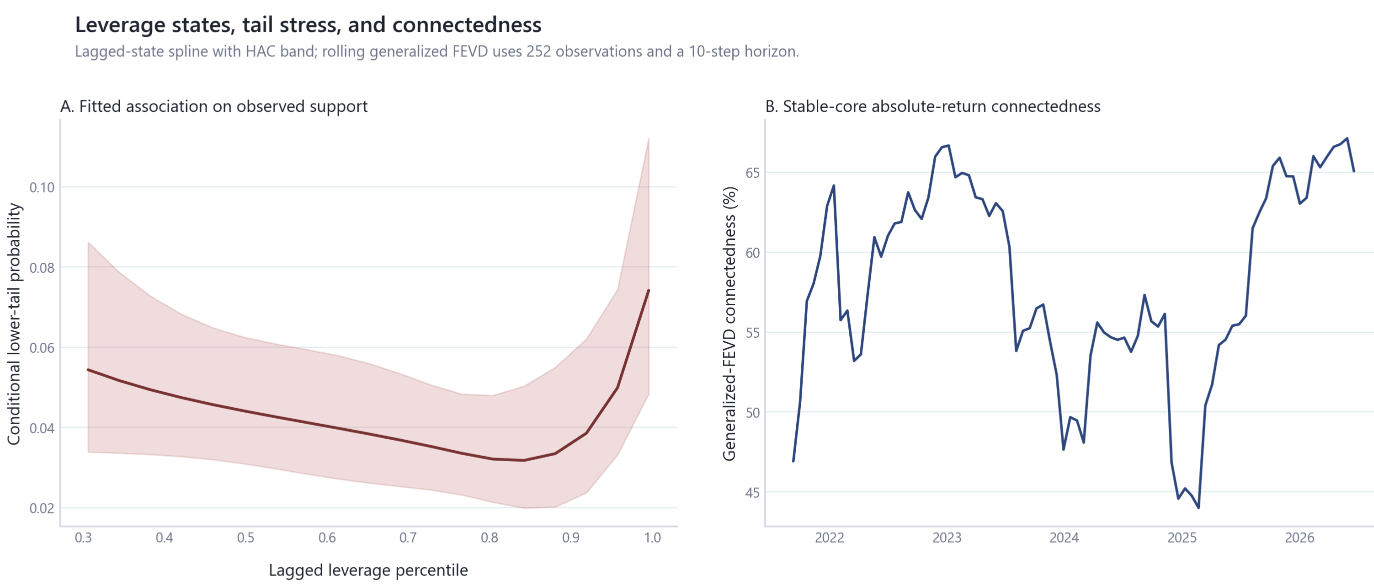
Sample. 2024-01-22 to 2026-04-10, n=426-557

Method and uncertainty. Distributed-lag HAC OLS with moving-block simultaneous bands; CFTC positioning is separate contemporaneous context. Intervals, diagnostics, and sensitivity specifications are reported in the linked source tables.

Evidence grade. B. **Limitation.** ETF reporting time is unresolved, CFTC data is weekly and released after the as-of date, and neither design identifies causality.

Provenance. [Primary source table](); [module documentation]().

03. Derivatives, Leverage, and Liquidations



Result. Across observed support, lagged BTC leverage states coincide with fitted next-session lower-tail probabilities from 3.2% to 7.4%.

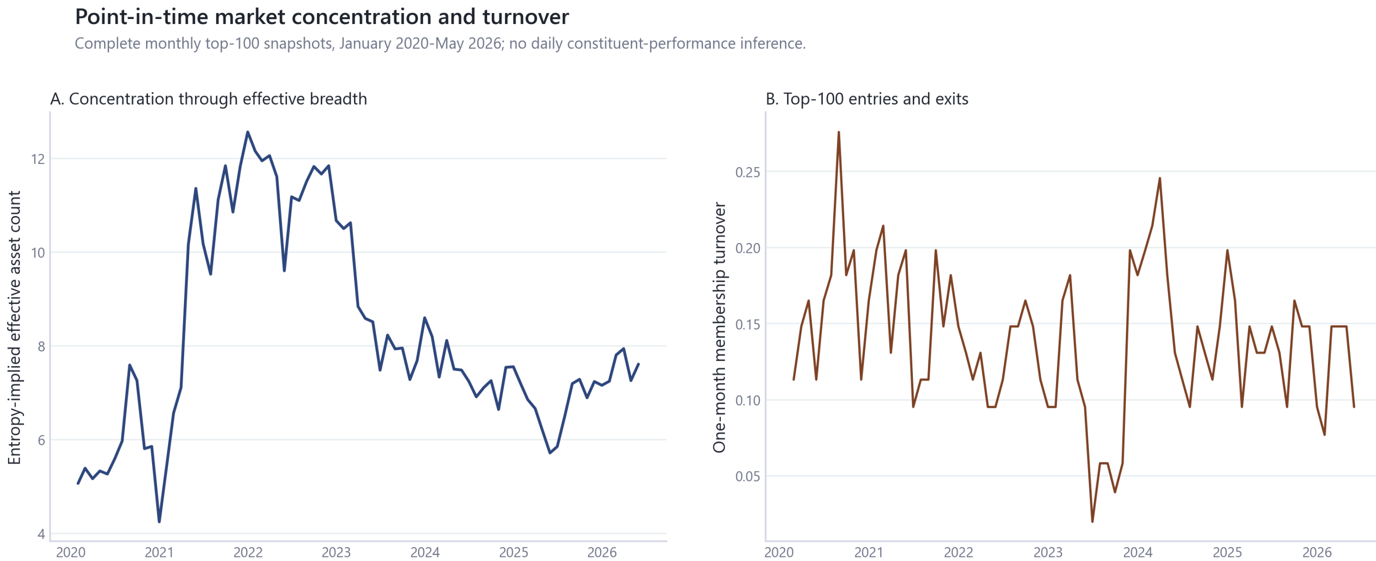
Sample. 2020-02-01 to 2026-04-12, n=2263

Method and uncertainty. Spline logistic association with HAC covariance, quantile/ES diagnostics, and generalized-FEVD connectedness. Intervals, diagnostics, and sensitivity specifications are reported in the linked source tables.

Evidence grade. B. **Limitation.** The relationship is associational, nonlinear, and not a forecast, trading signal, or liquidation-cause estimate.

Provenance. [Primary source table](); [module documentation]().

07. Chain Fundamentals and Sector Dynamics



Result. Across complete monthly PIT snapshots, effective asset count changed from 5.1 to 7.6, while median one-month membership turnover was 14.8%.

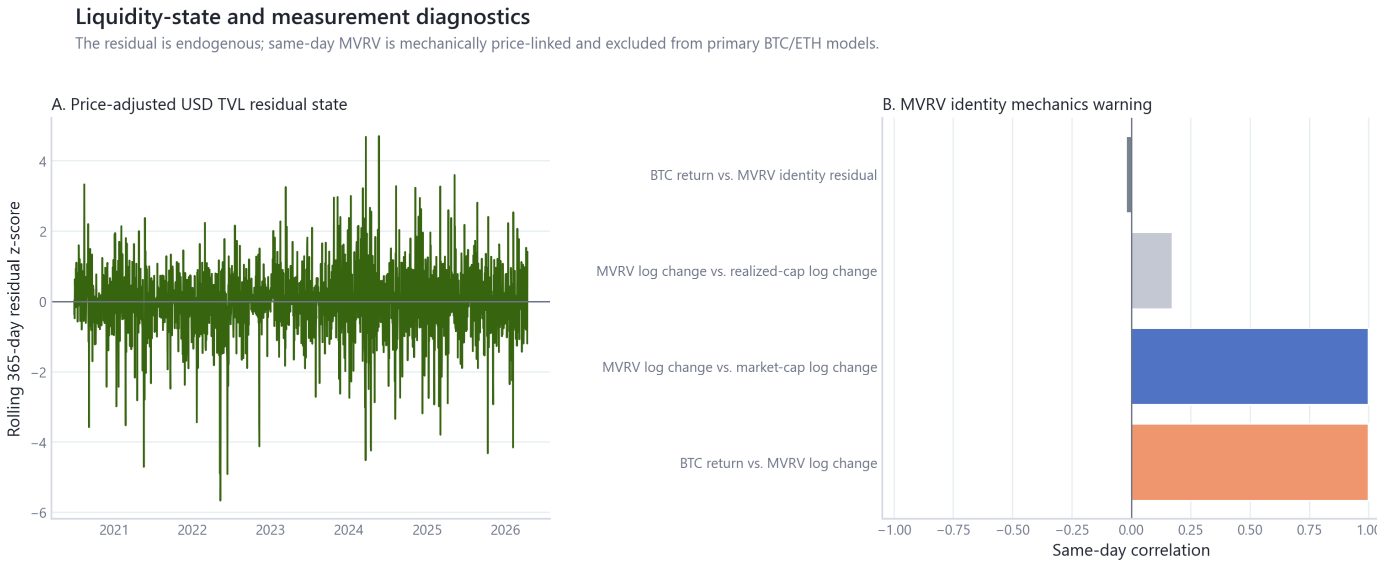
Sample. through 2026-05-31, result rows=77

Method and uncertainty. Monthly point-in-time concentration, entropy, effective count, membership transition, and exact HHI decomposition. Intervals, diagnostics, and sensitivity specifications are reported in the linked source tables.

Evidence grade. B. **Limitation.** Monthly snapshots support market-structure statements only; they do not establish daily constituent performance or historical altseason behavior.

Provenance. Primary source table; module documentation.

05. Stablecoin and DeFi Liquidity State



Result. Crypto-market return controls explain 0.4% of daily raw USD TVL growth in the stated HAC residualization; the remaining series is an endogenous liquidity-state proxy.

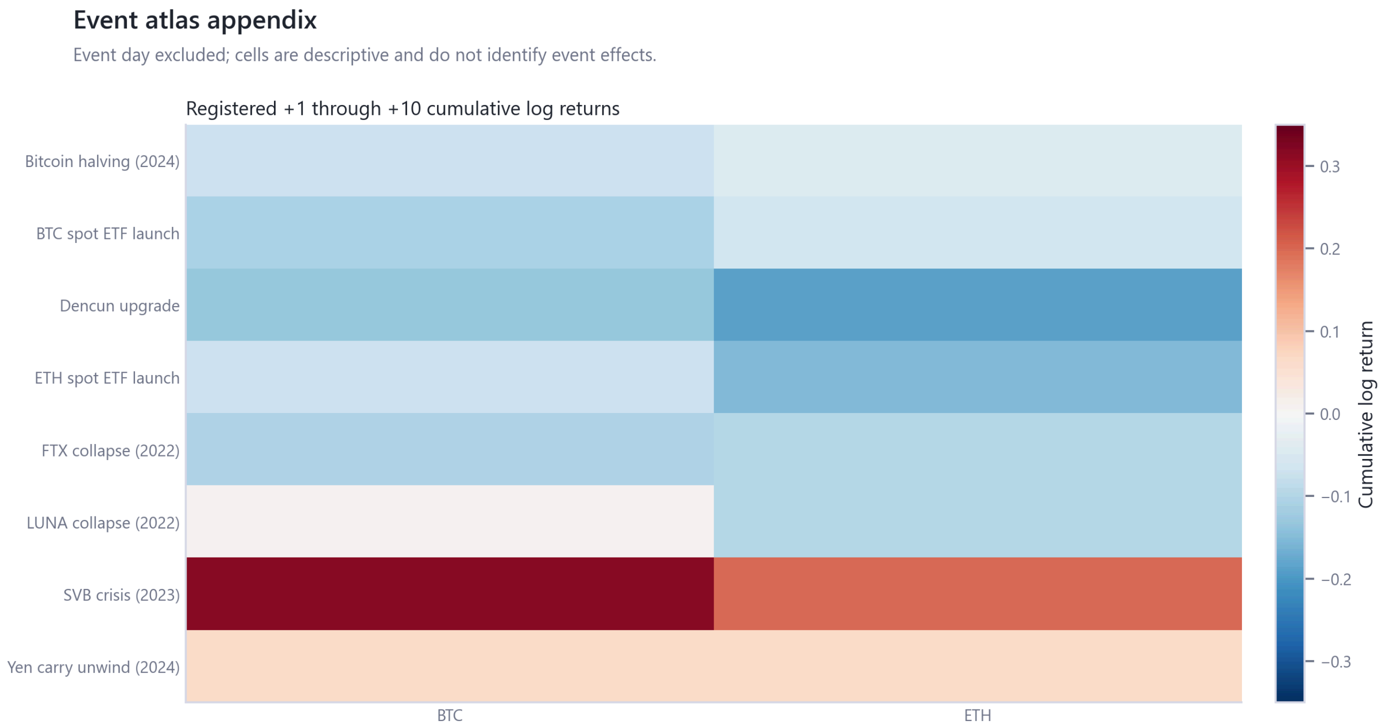
Sample. 2020-01-02 to 2026-04-14, n=2295

Method and uncertainty. HAC residualization of USD TVL growth; MVRV identity audit is appendix measurement evidence. Intervals, diagnostics, and sensitivity specifications are reported in the linked source tables.

Evidence grade. B. **Limitation.** USD TVL remains valuation-sensitive, stablecoin and TVL measures are endogenous, and same-day MVRV is mechanically price-linked.

Provenance. Primary source table; module documentation.

09. Event Stress and Cross-Module Synthesis



Result. Registered event-window responses vary substantially across events and assets and remain appendix sensitivity evidence rather than causal estimates.

Sample. 2020-01-02 to 2026-04-14, result rows=48

Method and uncertainty. Fixed post-event windows with empirical non-event-window comparison and cross-module claim ledger. Intervals, diagnostics, and sensitivity specifications are reported in the linked source tables.

Evidence grade. B. **Limitation.** Event windows are not an identification design; overlapping market developments and event selection constrain interpretation.

Provenance. Primary source table; module documentation.

Cross-Module Assessment

The strongest descriptive evidence concerns broad common crypto variation, lower-tail co-exceedance above independence, time-varying cross-market exposure, nonlinear leverage-state associations, and changing monthly market breadth and turnover. Formal TradFi era interactions are weak, most ETF lag coefficients do not clear simultaneous bands, and USD TVL residualization has low explanatory power. Those weak results are retained without specification search.

Reproducibility

```
uv sync --all-extras
uv run python scripts/run_all.py --mode local
uv run python scripts/execute_reproducibility_notebook.py
uv run python scripts/build_report.py
uv run python scripts/check_research_surface.py --module all
```

Public CI runs fixture scientific smoke tests and validates committed semantic outputs without requiring local provider exports. A full local build requires legally obtained inputs under `data_local/raw/` or an external read-only `CMD_DATA_ROOT`.

References

Source attribution, event provenance, and econometric references are maintained in `REFERENCES.md`. Source eligibility and fallback decisions are in `research/source_decisions.csv`.